

Malignant Infarction

Decompressive hemicraniectomy selection criteria, evidence, and supportive ICU care.

STAGE 1: 0 - 24 HOURS

Baseline Core & Serial NIHSS Checks

STAGE 2: 24 - 48 HOURS

Peak Edema Phase & Serial CT Scan

STAGE 3: < 48H SURGERY

Decompressive Hemicraniectomy

1. Decompressive Hemicraniectomy Selection Criteria

Clinical Deficit Severity:

- NIHSS > 15 (non-dominant hemisphere)
- NIHSS > 20 (dominant hemisphere)
- AND decrease in level of consciousness (NIHSS Item 1a score ≥ 1 / obtunded or stuporous)
- **Timing:** Surgery performed **within 48 hours** of onset.

Radiographic Markers:

- Infarction of $\geq 50\%$ of the MCA territory (CT/MRI)
- DWI core volume > 82 mL within 6 hours
- DWI core volume > 145 mL within 14 hours
- Midline shift or mass effect on repeat imaging
- **Surgical Spec:** Bone flap diameter $\geq 12\text{--}15$ cm with duraplasty.

2. Surgical Outcomes & Evidence (By Age Group)

Age 18–60 Years (pooled DECIMAL/DESTINY/HAMLET)
Surgery (22% Mort) vs Med (71% Mort)



Age 18–60 Medical Control



Age ≥ 61 Years (DESTINY II)
Surgery (33% Mort) vs Med (70% Mort)



Age ≥ 61 Medical Control



■ mRS 0–2: Functional independence; mRS 3: walks unassisted but needs some help

■ mRS 4: Moderately severe; unable to walk or attend bodily needs unassisted

■ mRS 5: Severe disability; bedridden / constant care ■ mRS 6: Death

- **Age 18–60:** NNT = 2 for survival, NNT = 4 for survival with mRS ≤ 3 (able to walk unassisted). | • **Age ≥ 61 :** NNT = 3 for survival, NNT = 25 for mRS ≤ 3 . *Goals-of-care discussion critical.

3. Supportive ICU Care & Medical Management

Positioning	Elevate HOB 30 degrees; maintain straight head/neck alignment to maximize venous outflow.
Fluids	Maintain euvolemia with isotonic fluids. Avoid hypotonic fluids (e.g. D5W, 0.45% NS) that can worsen edema; balanced crystalloids such as LR should follow local neuro-ICU protocol.
Osmotherapy	Consider targeted PRN hyperosmolar agents (HTS 3% or Mannitol) for acute decline or severe mass effect. <i>Prophylactic osmotherapy is not recommended.</i>
Metabolic	Target normothermia (<37.8°C). Target normocapnia (PaCO ₂ 35-45 mmHg); avoid hypoventilation/hypercapnia.
Steroids	Class III (Harmful): Corticosteroids are NOT recommended for reducing cerebral edema in acute ischemic stroke.

DECIMAL: Vahedi K et al. *Stroke*. 2007;38:2506-2517. [PMID: 17690311](#) | **DESTINY:** Jüttler E et al. *Stroke*. 2007;38:2518-2525. [PMID: 17690310](#)
HAMLET: Hofmeijer J et al. *Lancet Neurol*. 2009;8:326-333. [PMID: 19269254](#) | **DESTINY II:** Jüttler E et al. *N Engl J Med*. 2014;370:1091-1100. [PMID: 24645942](#)
AHA Guidelines: Wijdicks EF et al. *Stroke*. 2014;45:1222-1238. [PMID: 24481970](#)